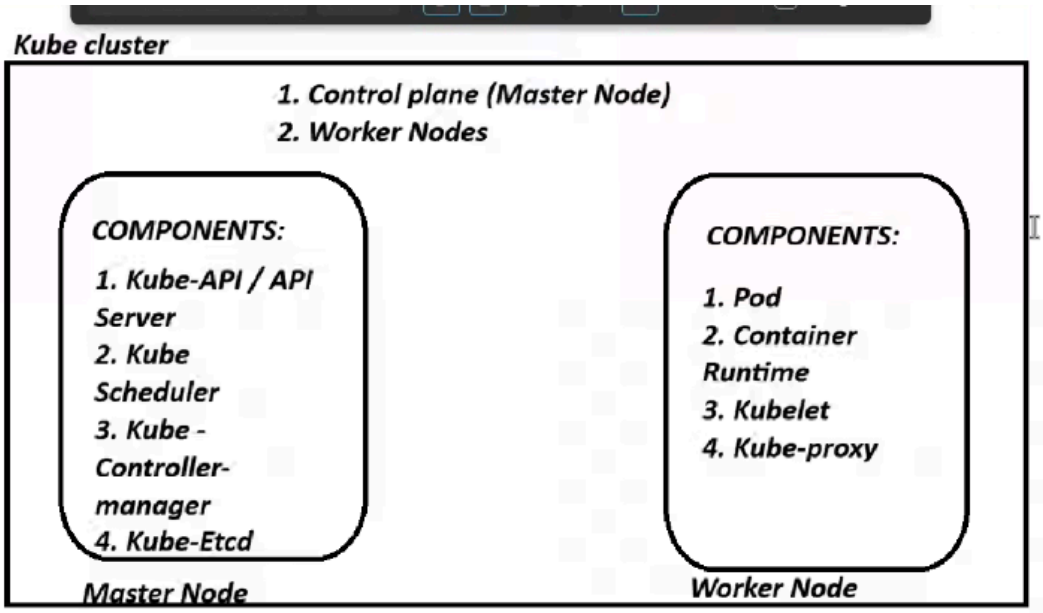
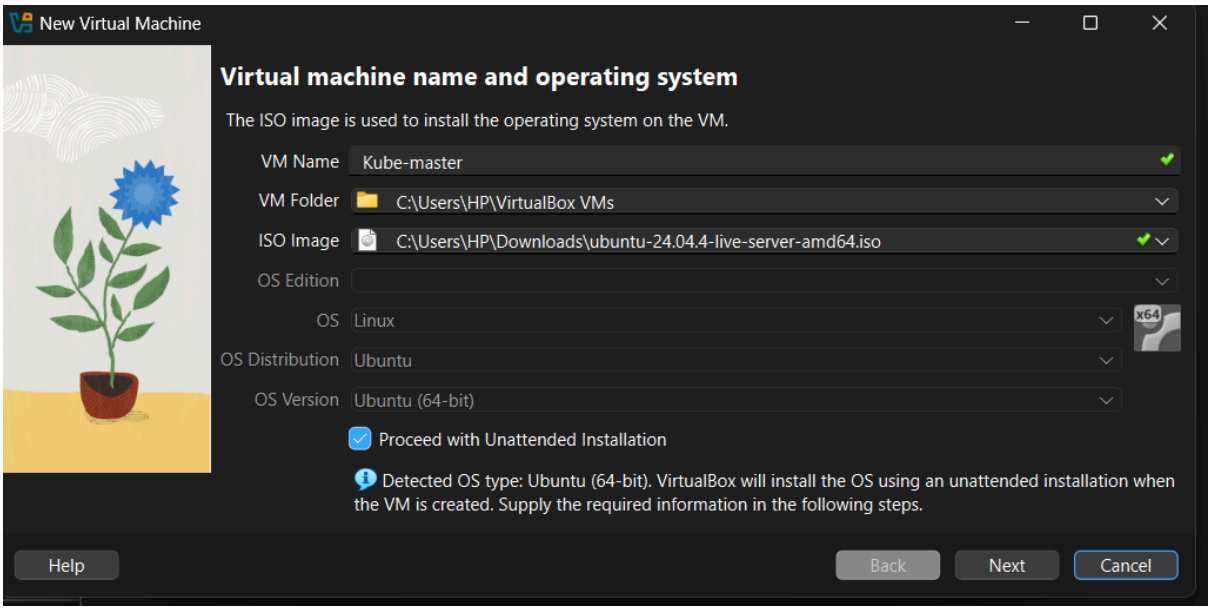
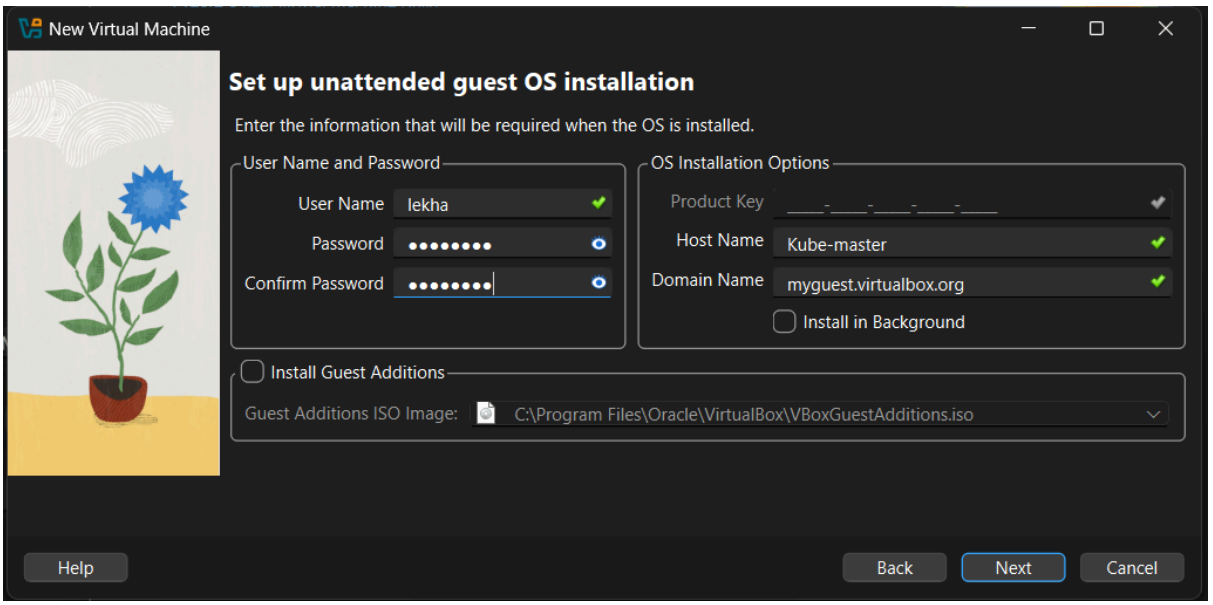
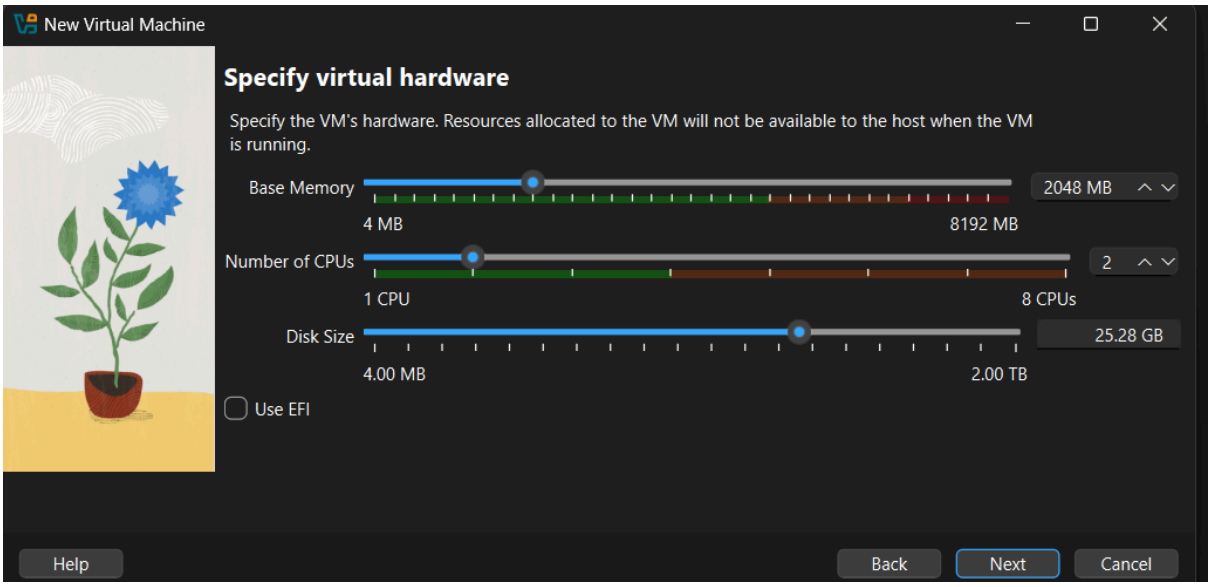
DEVEOPS - 05-03-2026

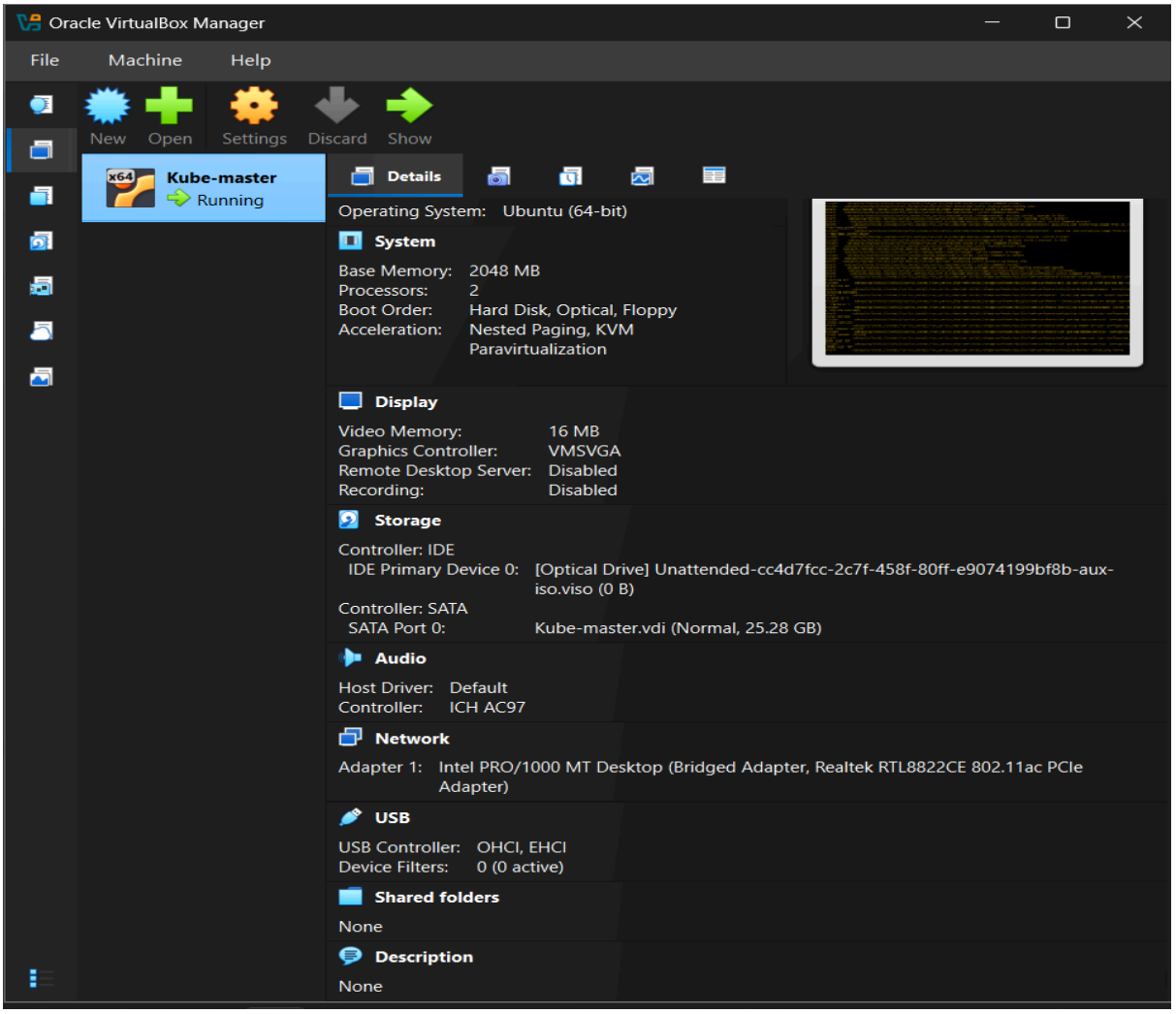
Kubernetes

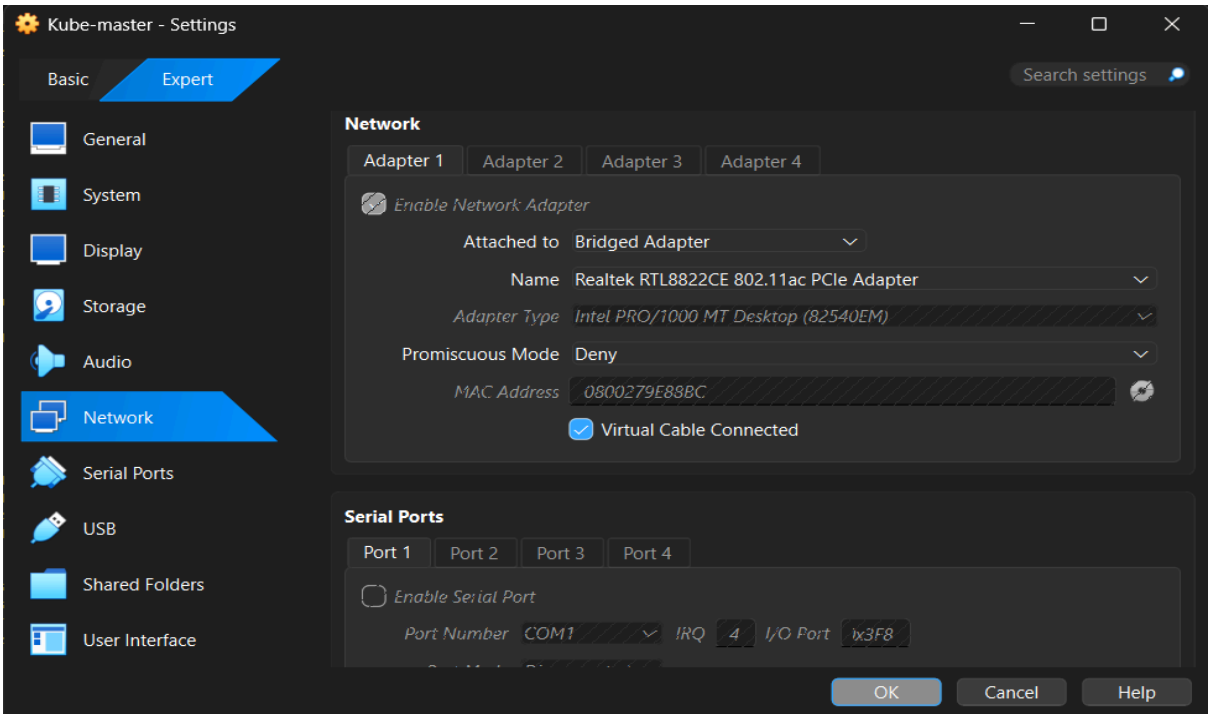


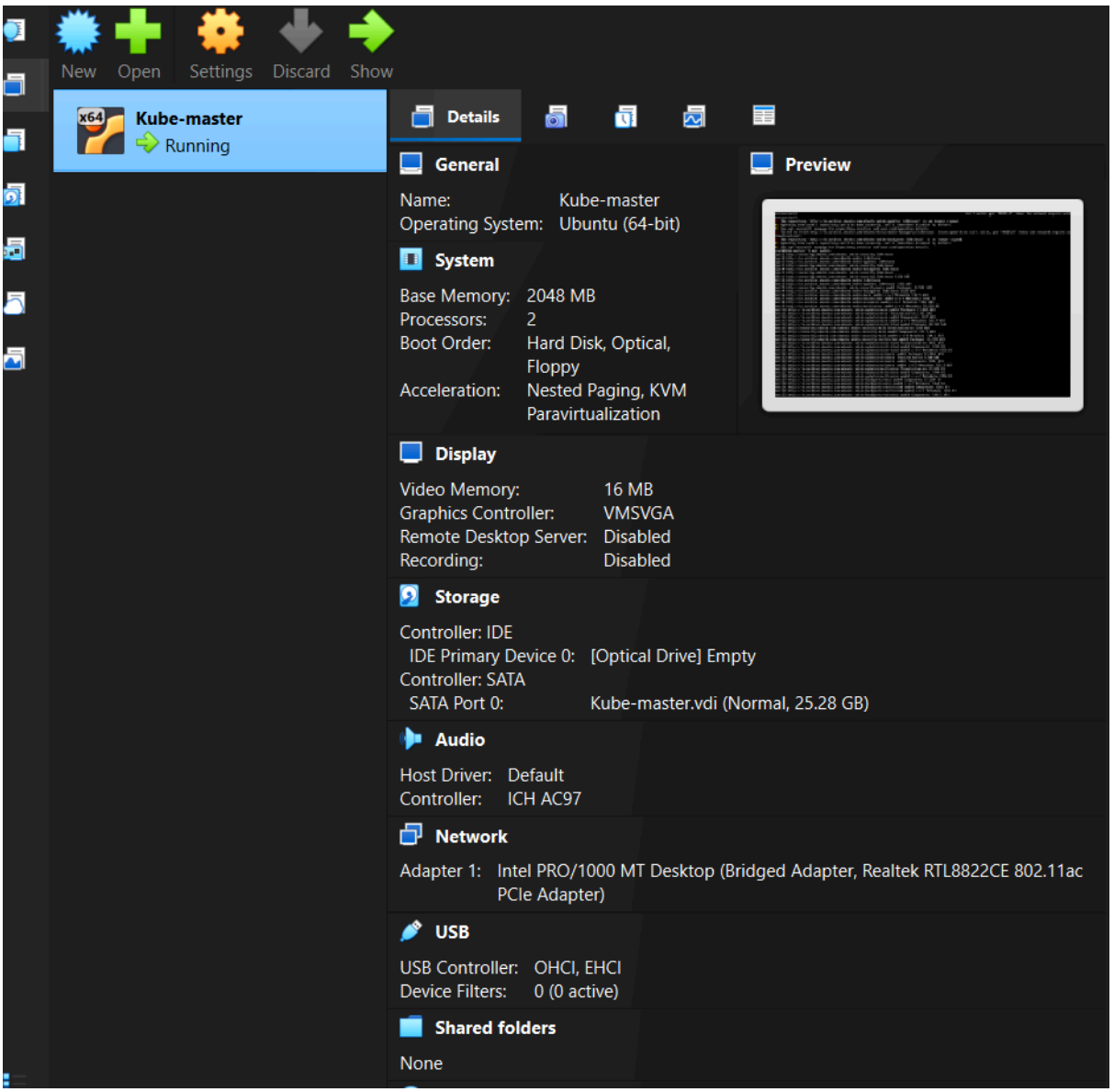



New Virtual Machine
Set up unattended guest OS installation
Enter the information that will be required when the OS is installed.
User Name and Password
User Name
lekha
Password
Confirm Password
OS Installation Options
Product Key
Host Name
Kube-master
Domain Name
myguest.virtualbox.org
Install in Background
Install Guest Additions
Guest Additions ISO Image:
C:\Program Files\Oracle\VirtualBox\VBoxGuestAdditions.iso
Help
Back
Next
Cancel


New Virtual Machine
Specify virtual hardware
Specify the VM's hardware. Resources allocated to the VM will not be available to the host when the VM is running.
Base Memory
2048 MB
4 MB
8192 MB
Number of CPUs
2
1 CPU
8 CPUs
Disk Size
25.28 GB
4.00 MB
2.00 TB
Use EFI
Help
Back
Next
Cancel


Oracle VirtualBox Manager
File
Machine
Help
New
Open
Settings
Discard
Show
Kube-master
Running
Details
Operating System: Ubuntu (64-bit)
System
Base Memory: 2048 MB
Processors: 2
Boot Order: Hard Disk, Optical, Floppy
Acceleration: Nested Paging, KVM Paravirtualization
Display
Video Memory: 16 MB
Graphics Controller: VMSVGA
Remote Desktop Server: Disabled
Recording: Disabled
Storage
Controller: IDE
IDE Primary Device 0: [Optical Drive] Unattended-cc4d7fcc-2c7f-458f-80ff-e9074199bf8b-aux-iso.viso (0 B)
Controller: SATA
SATA Port 0: Kube-master.vdi (Normal, 25.28 GB)
Audio
Host Driver: Default
Controller: ICH AC97
Network
Adapter 1: Intel PRO/1000 MT Desktop (Bridged Adapter, Realtek RTL8822CE 802.11ac PCIe Adapter)
USB
USB Controller: OHCI, EHCI
Device Filters: 0 (0 active)
Shared folders
None
Description
None


Kube-master - Settings
Basic
Expert
Search settings
General
System
Display
Storage
Audio
Network
Serial Ports
USB
Shared Folders
User Interface
Network
Adapter 1
Adapter 2
Adapter 3
Adapter 4
Enable Network Adapter
Attached to Bridged Adapter
Name Realtek RTL8822CE 802.11ac PCIe Adapter
Adapter Type Intel PRO/1000 MT Desktop (82540EM)
Promiscuous Mode Deny
MAC Address 0800279E88BC
Virtual Cable Connected
Serial Ports
Port 1
Port 2
Port 3
Port 4
Enable Serial Port
Port Number COM1
IRQ 4
I/O Port 0x3F8
OK
Cancel
Help



Kube-master is running, then the oracle virtualbox will open
Give username and password

Then sudo -i => to change as root user
apt -update => update it
apt install openshh-server -y
ip a => to view ip address

Find ip address and put in windows powershell

```
PS C:\Users\HP> ssh lekha@10.1.10.144
lekha@10.1.10.144's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-100-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Thu Mar  5 09:22:11 AM UTC 2026

  System load:  0.03               Processes:               121
  Usage of /:   10.8% of 24.71GB   Users logged in:         1
  Memory usage: 12%                IPv4 address for enp0s3: 10.1.10.144
  Swap usage:   0%


Expanded Security Maintenance for Applications is not enabled.

31 updates can be applied immediately.
25 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Che


Last login: Thu Mar  5 09:13:20 2026 from 10.1.10.129
lekha@Kube-master:~$
```

After this the oracle virtualbox can be access from terminal
sudo -i

apt update

```
apt install -y apt-transport-https ca-certificates curl gpg
```

```
mkdir -p /etc/apt/keyrings
```

```
curl -fsSL https://pkgs.k8s.io/core:/stable:/v1.29/deb/Release.key | gpg --dearmor -o /etc/apt/keyrings/kubernetes-apt-keyring.gpg
```

```
echo "deb [signed-by=/etc/apt/keyrings/kubernetes-apt-keyring.gpg] https://pkgs.k8s.io/core:/stable:/v1.29/deb/ /" | tee /etc/apt/sources.list.d/kubernetes.list
```

```
sudo apt install -y containerd
```

```
sudo mkdir -p /etc/containerd

sudo containerd config default | sudo tee /etc/containerd/config.toml

sudo systemctl restart containerd

sudo systemctl enable containerd

apt update

apt install -y kubelet kubeadm kubectl

apt-mark hold kubelet kubeadm kubectl

sudo swapoff -a

free -m
```

vmware power off

Clone pannum pothu every details like name, password, ip address, machine id… everything will be same so same location la koodutha overwrite agum so change the location

right click => manage => clone => setup open agum => next => next => create a full clone => next => name: kube-worker => location: not old vm location some other=> finish => close.

Then it will be cloned properly.

Oracle la exit power off => right click => clone => change name and location => mac address: generate new mac address for all => next => full clone => finish